

Zhenkai Zhang (Ken)

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EDUCATIONAL BACKGROUND

The Hong Kong Polytechnic University

Starting Sep 2026

MSc in Data Science & Analytics

Nanjing Tech University

Sep 2022 - Jun 2026

BSc in Data Science and Big Data Technology | 3.9/4.0

RESEARCH INTERESTS

Efficient AI; model reuse and routing; generative AI; multimodal agents

SELECTED PUBLICATIONS

Zhang, Z., Gao, Y., & Chen, D. (2025). Exploring and Enhancing Advanced MoE Models: From DeepSpeed-MoE to DeepSeek-V3.

2025 IEEE 6th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT). First author. [DOI](#)

Zhang, Z., Ma, T., Yao, Y., et al. (2025). Predicting Olympic Medal Performance for 2028: Machine Learning Models and the Impact of Host and Coaching Effects. Applied Sciences, 15(14), 7793. First author. [DOI](#)

Xia, W., Gao, Y., **Zhang, Z.**, et al. (2025). Behavior Prediction of Connections in Eco-Designed Thin-Walled Steel-Ply-Bamboo Structures Based on Machine Learning for Mechanical Properties. Sustainability, 17(15), 6753. Third author. [DOI](#)

SELECTED PROJECTS

RouterBench-Mini: Cost-Aware Model Reuse for Multimodal Agents

Jul 2026

Independent Research Project | [GitHub](#)

- Designed and implemented a cost-aware benchmark for routing between Qwen3.5-35B-A3B and Qwen3.5-397B-A17B across 600 text, vision, and tool-use tasks.
- Compared task-aware, confidence-based, and learned quality-gap routing with fingerprint-disjoint evaluation, calibration, ablations, paired bootstrap confidence intervals, and failure analysis.
- Across two non-overlapping evaluation batches, task-aware routing trailed Always-Strong by 0.67 percentage points while reducing average cost by 22.5% and latency by 26.6%; released reproducible code, results, tests, and CI.

Finance-Agent: Multi-Agent Financial Analysis System

Oct 2025 - Jan 2026

Engineering Project | [GitHub](#)

- Designed and implemented an MCP-based multi-agent investment advisory system for structured analysis and recommendation generation across 4,000+ China A-share stocks, addressing data inconsistency and incomplete analytical coverage in general-purpose LLM systems.
- Collected and annotated approximately 100,000 financial news articles and fine-tuned Qwen3-2.5B with LoRA for sentiment classification and risk prediction, achieving test accuracies of 91% and 88%, respectively.
- Standardized financial APIs as MCP tools with a unified schema, achieving a 98% tool-call success rate and approximately 99% data consistency.
- Built four specialized agents for fundamental, technical, valuation, and news analysis using LangGraph and ReAct, together with a synthesis agent for generating structured Markdown reports.

TECHNICAL SKILLS

Python, PyTorch, Transformers, scikit-learn, PEFT/LoRA, LangGraph, MCP, Git, Linux, pytest

HONORS

Meritorious Winner (Problem F), Mathematical Contest in Modeling (MCM)

Feb 2026